Data_Analyst — Step 2

Learn Statistics

Detailed Learning Notes • CareerCompass

1. Learning Objective

Develop enough statistical reasoning to summarize business data, compare groups and communicate uncertainty responsibly.

2. Core Concepts — Detailed Breakdown

Descriptive statistics

Use mean, median, percentiles, variance and standard deviation to summarize distributions.

Sampling

Understand population, sample, bias and sampling variability.

Comparisons

Use confidence intervals and hypothesis tests conceptually to compare groups.

Relationships

Interpret correlation and simple regression while recognizing confounding variables.

Business interpretation

Translate statistical output into plain-language conclusions and clearly state limitations.

3. Recommended Learning Workflow

- Practice descriptive statistics.
- Visualize distributions.
- Run simple sampling simulations.
- Study confidence intervals and hypothesis tests.
- Apply the concepts to a business dataset.

5. Hands-on Project / Practice

Compare conversion rates between two campaign groups using an appropriate method. Report sample sizes, effect size, uncertainty and a plain-language recommendation rather than only a p-value.

6. Common Mistakes & Pitfalls

- Treating $p < 0.05$ as automatic business success.
- Ignoring sample size.
- Confusing statistical and practical significance.
- Using inappropriate averages for skewed data.
- Assuming observational comparisons are causal.

7. Completion Checklist

- I can interpret common descriptive statistics.
- I can explain sampling variability.
- I can interpret a confidence interval.
- I can distinguish correlation from causation.
- I can communicate statistical results clearly.

8. Interview / Assessment Questions

- Mean vs median?
- What is a confidence interval?
- What does a p-value mean?
- Why does sample size matter?
- Why does correlation not imply causation?

9. Recommended References

Khan Academy Statistics & Probability: https://www.khanacademy.org/statistics-probability

Data Science Tutorial: https://www.youtube.com/watch?v=ua-CiDNNj30

Note: These notes are designed as a structured learning guide. Use the references for deeper, current documentation and hands-on practice.